

Exercises 9.A

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Exercise 3. Suppose V is a real vector space and $v_1, \dots, v_m \in V$. Prove that $v_1, \dots, v_m$ is linearly independent in V_C if and only if $v_1, \dots, v_m$ is linearly independent in V .

Proof. First assume that $v_1, \dots, v_m$ is linearly independent in V_C . If $v_1, \dots, v_m$ was linearly dependent in V , we would have

$$c_1 v_1 + \dots + c_m v_m = 0 \quad (1)$$

for some $c_1, \dots, c_m \in \mathbb{R}$ not all zero. Then considering (1) in V_C produces a contradiction.

For the second part, assume that $v_1, \dots, v_m$ are linearly dependent in V_C . Then we have

$$c_1 v_1 + \dots + c_m v_m = 0 \quad (2)$$

for some $c_1, \dots, c_m \in \mathbb{C}$ not all zero. For each c_j , write $c_j = a_j + b_j i$ where $a_j, b_j \in \mathbb{R}$. Then (2) becomes

$$\begin{aligned} 0 &= (a_1 + b_1 i)v_1 + \dots + (a_m + b_m i)v_m \\ &= a_1 v_1 + b_1 i v_1 + \dots + a_m v_m + b_m i v_m \\ &= (a_1 v_1 + \dots + a_m v_m) + i(b_1 v_1 + \dots + b_m v_m). \end{aligned}$$

Then both the "real" and "imaginary" parts of the vector in V_C from the last equation must be equal to zero. In particular,

$$a_1 v_1 + \dots + a_m v_m = 0 \quad (3)$$

and

$$b_1 v_1 + \dots + b_m v_m = 0 \quad (4)$$

If $a_1, \dots, a_m$ are all zero, then $b_1, \dots, b_m$ are not all zero since $c_1, \dots, c_m$ are not all zero, so (4) is a linear combination of $v_1, \dots, v_m$ with real coefficients which are not all zero, and thus $v_1, \dots, v_m$ is linearly dependent in V . If $a_1, \dots, a_m$ are not all zero, then (3) produces such a combination, so we are done. $\square$

Exercise 6. Suppose V is a real vector space and $T \in \mathcal{L}(V)$. Prove that T_C is invertible if and only if T is invertible.

Proof. If T_C is invertible, then for any nonzero $v \in V$,

$$\begin{aligned} T_C(v + i0) &= Tv + iT0 \\ &= T(v) \\ &\neq 0, \end{aligned}$$

so T is invertible.

If T is invertible, then given nonzero $u + iv \in V_C$, we have

$$T_C(u + iv) = Tu + iTv.$$

Since T is invertible and $u, v \in V$ are not both zero, Tu and Tv are also not both zero, and thus $T_C(u + iv)$ is nonzero. $\square$

Exercise 9. Prove there does not exist an operator $T \in \mathcal{L}(\mathbb{R}^7)$ such that $T^2 + T + I$ is nilpotent.

Proof. If $T^2 + T + I$ is nilpotent, then $(T^2 + T + I)^n = 0$ for some n . This would mean that 0 is the only eigenvalue of $T^2 + T + I$, and in particular that $(T^2 + T + I)v = 0$ for some nonzero $v \in V$. Consider this polynomial applied to T_C , and after factoring it we get

$$\left(T_C + \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) I \right) \left(T_C + \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i \right) I \right) v = 0,$$

so either $-\frac{1}{2} - \frac{\sqrt{3}}{2}i$ or $-\frac{1}{2} + \frac{\sqrt{3}}{2}i$ is an eigenvalue of T_C . Of course, since neither is real, neither can be an eigenvalue of T , so both

$$\left(T_C + \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i \right) I \right) w$$

and

$$\left(T_C + \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i \right) I \right) w$$

are nonzero for any $w \in V$. Thus

$$(T^2 + T + I)w = (T_C^2 + T_C + I)w \neq 0,$$

contradicting that 0 is an eigenvalue of $T^2 + T + I$.

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Exercise 10. Give an example of an operator $T \in \mathcal{L}(\mathbb{C}^7)$ such that $T^2 + T + I$ is nilpotent.

Proof. Let $\lambda = -\frac{1}{2} - \frac{\sqrt{3}}{2}i$ and define T by

$$T(z_1, z_2, z_3, z_4, z_5, z_6, z_7) = (\lambda z_1, \lambda z_2, \lambda z_3, \lambda z_4, \lambda z_5, \lambda z_6, \lambda z_7).$$

Clearly $T^2 + T + I$ sends each coordinate z_j to $(\lambda^2 + \lambda + 1)z_j$. Note that $\lambda^2 = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$, so

$$\begin{aligned}\lambda^2 + \lambda + 1 &= \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) + \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right) + 1 \\ &= \left(-\frac{1}{2} - \frac{1}{2}\right) + \left(\frac{\sqrt{3}}{2}i - \frac{\sqrt{3}}{2}i\right) + 1 \\ &= -1 + 0 + 1 \\ &= 0,\end{aligned}$$

so in fact each coordinate gets sent to 0. Thus $T^2 + T + I = 0$, so it is nilpotent. $\square$